# Generalized Linear Models

## Binary outcomes and logistic regression

Foundations of Regression and Predictive Modeling

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# Generalized linear models

So far, our response variable has usually been numerical:
*price*, *hours*, *minutes*, *demand*, *delay.*

But many practical questions have a different kind of response:

- did the home team win?
- did a passenger make the connection?
- how many trips started here?
- how many support tickets arrived?

Today: binary outcomes.

# A first binary question

Sportsbooks publish a point spread for NBA games.

In this data set:

- `HomeSpread` is the spread in favor of the home team,
- negative means the visitor is favored,
- positive means the home team is favored,
- `HomeWin` records whether the home team won.

Question:

*How does the point spread translate into home-team win probability?*

# A few rows

| Game | Season | Home spread | Result |
|---|---|---|---|
| 81 | 2021 | -8.4 | Visitor win |
| 436 | 2024 | 3.8 | Home win |
| 911 | 2020 | 12.1 | Home win |
| 1240 | 2023 | -1.9 | Home win |
| 1557 | 2022 | 6.7 | Visitor win |
| 2016 | 2025 | 0.4 | Home win |

The response is a 0/1 event.

That changes the modeling problem.

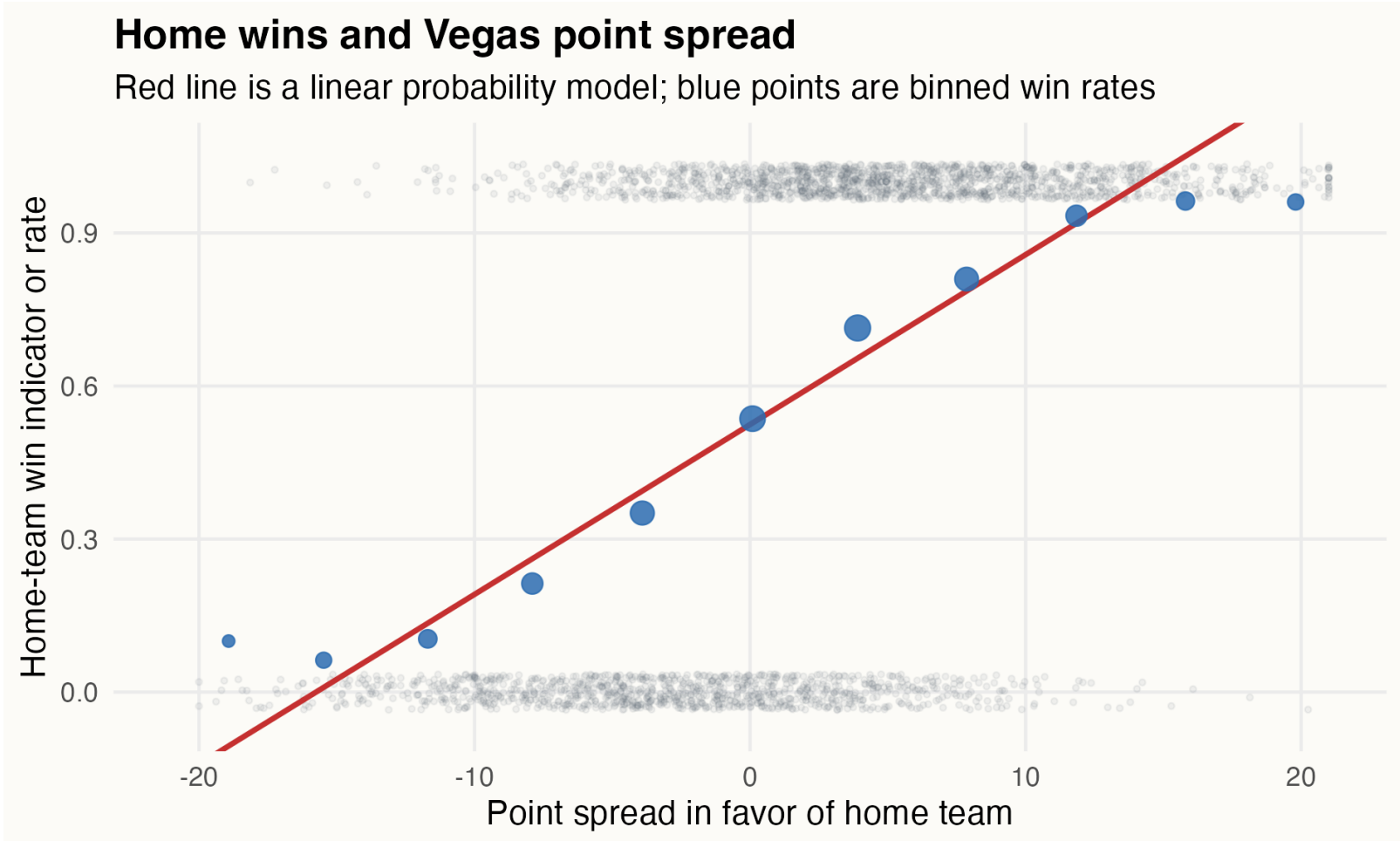
# First try

Could we fit ordinary regression?

$$\text{HomeWin}_i = \beta_0 + \beta_1 \text{HomeSpread}_i + e_i.$$

This is called a **linear probability model**.

It is tempting because the fitted value looks like a probability.


Home wins and Vegas point spread
Red line is a linear probability model; blue points are binned win rates
Home-team win indicator or rate
0.0
0.3
0.6
0.9
-20
-10
0
10
20
Point spread in favor of home team

# What goes wrong?

A probability has to live between 0 and 1.

A straight line does not know that.

It can predict impossible probabilities if we ask about large enough spreads.

And the relationship is not really straight. Moving from -12 to -8 matters differently than moving from 0 to 4.

# Model the odds

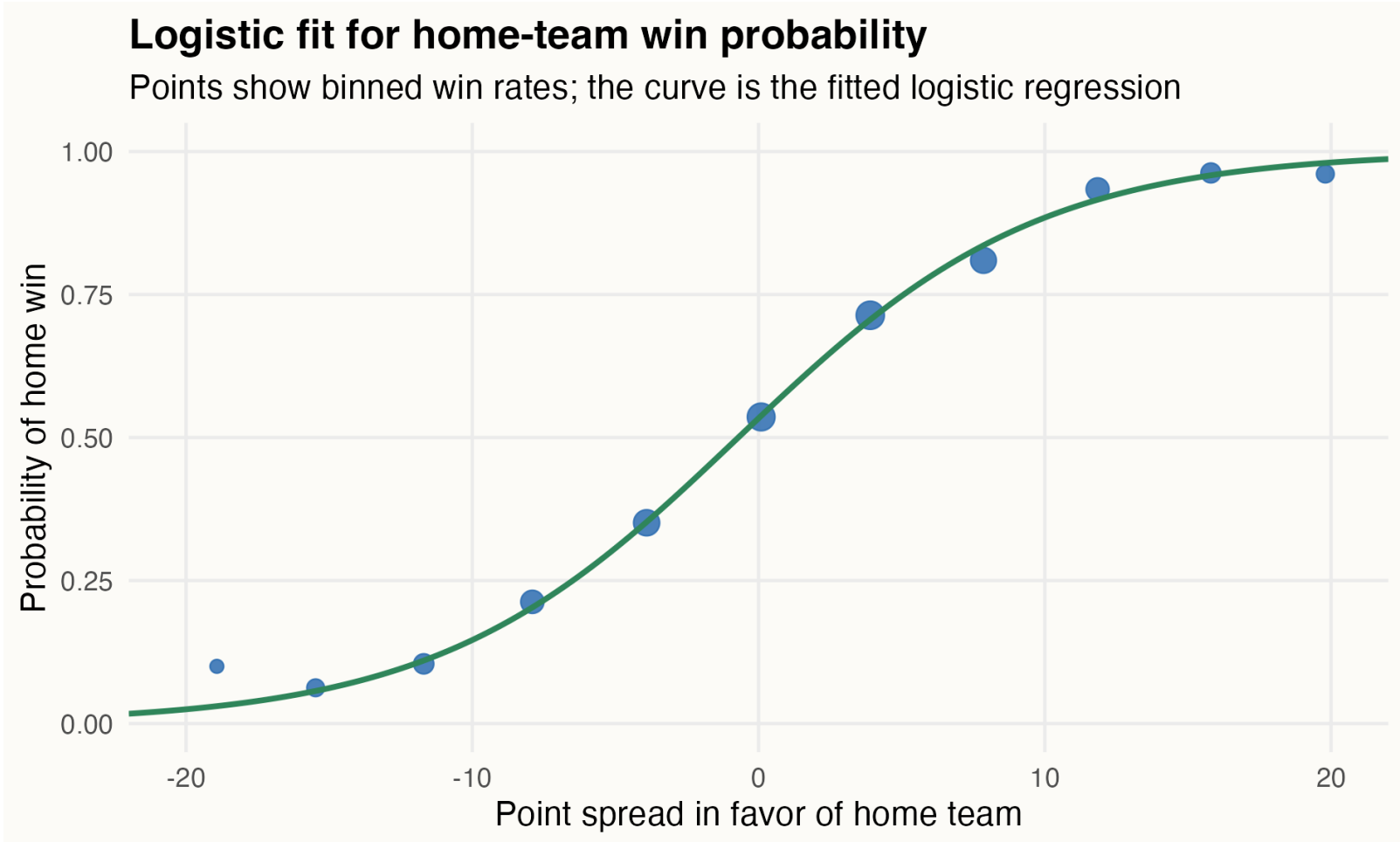
Instead of modeling probability directly, logistic regression models log odds.

If $p_i$ is the probability of a home win:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 \text{HomeSpread}_i.$$

The left side can be any real number.

The probability still stays between 0 and 1.


Logistic fit for home-team win probability
Points show binned win rates; the curve is the fitted logistic regression
Probability of home win
1.00
0.75
0.50
0.25
0.00
-20
-10
0
10
20
Point spread in favor of home team

# In R

```
nba_fit <- glm(
  HomeWin ~ HomeSpread,
  data = nba_games,
  family = binomial()
)
```

The new part is:

```
family = binomial()
```

That tells R we have a binary response and want the logit link.

# Reading the coefficient

The fitted model is approximately:

$$\log\left(\frac{\hat{p}}{1-\hat{p}}\right) = 0.135 + 0.190 \cdot \text{HomeSpread}.$$

The slope is on the log-odds scale.

One more point of home-team spread adds `0.190` to the log odds of a home win.

# Odds ratios

Exponentiate the slope:

$$e^{0.190} \approx 1.21.$$

So:

*each additional point of home-team spread multiplies the odds of a home win by about 1.21.*

That is an odds ratio.

# Probabilities are not additive

The same odds ratio can imply different probability changes.

In this model:

- home team favored by 10: fitted win probability about `0.88`,
- pick'em game: fitted win probability about `0.53`,
- visitor favored by 10: fitted win probability about `0.15`.

The coefficient is constant on the log-odds scale.

It is not a constant probability difference.

# More than one input

Now a more operational binary problem.

An airline studies passenger connections through six major hubs:

- ATL, CLT, DFW,
- DEN, ORD, IAH.

Each row is one passenger connection.

Question:

*Which connections are likely to be made?*

# The data

For each connection:

- `MadeConnection`: made or missed,
- `ScheduledConnectionMinutes`: scheduled time between flights,
- `ArrivalDelayMinutes`: arrival delay of the inbound flight,
- `Hub`: connecting airport,
- `TerminalChange`: whether the passenger had to change terminals.

This is not just an airline problem.

It is the same modeling problem as churn, conversion, default, fraud, and admission.

# A few rows

| Hub | Scheduled connection | Arrival delay | Terminal change | Result |
|---|---|---|---|---|
| DEN | 155 | -15 | Same terminal | Made connection |
| DFW | 52 | 16 | Same terminal | Missed connection |
| CLT | 118 | -6 | Same terminal | Made connection |
| DFW | 59 | -2 | Terminal change | Made connection |
| DFW | 35 | -13 | Same terminal | Made connection |
| CLT | 67 | 0 | Same terminal | Made connection |

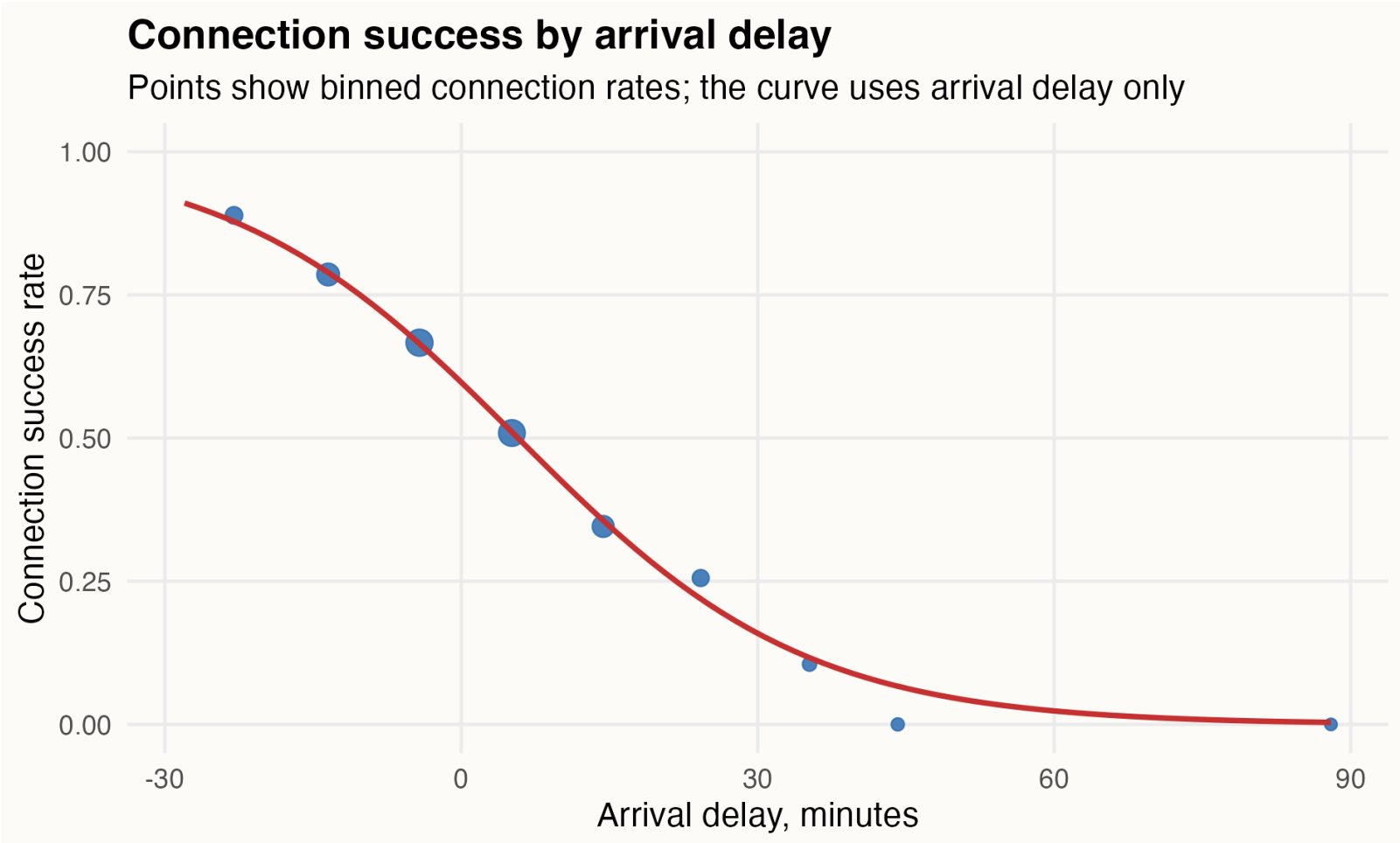
Already: delay alone cannot be the whole story.

# Start with arrival delay

First model:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 \text{ArrivalDelayMinutes}_i.$$

Here $p_i$ is the probability that passenger $i$ makes the connection.


Connection success by arrival delay
Points show binned connection rates; the curve uses arrival delay only
Connection success rate
1.00
0.75
0.50
0.25
0.00
-30
0
30
60
90
Arrival delay, minutes

# Delay is not enough

Arrival delay matters.

But a 20-minute delay is not the same thing for every passenger.

It depends on how much connection time the itinerary had in the first place.

That is the next input.

# Add connection time

The model becomes:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 \text{ScheduledConnectionMinutes}_i + \beta_2 \text{ArrivalDelayMi}$$

The comparison:

*scheduled connection time, holding arrival delay fixed.*

and

*arrival delay, holding scheduled connection time fixed.*

# Interpreting ten minutes

The adjusted model estimates:

- 10 more scheduled connection minutes multiply the odds of making the connection by about `2.15`,
- 10 more minutes of arrival delay multiply the odds by about `0.38`.

In words:

*more scheduled buffer helps; later arrival hurts.*

No surprise. But now we can quantify both in the same model.

# Terminal changes

A terminal change is not the same at every airport.

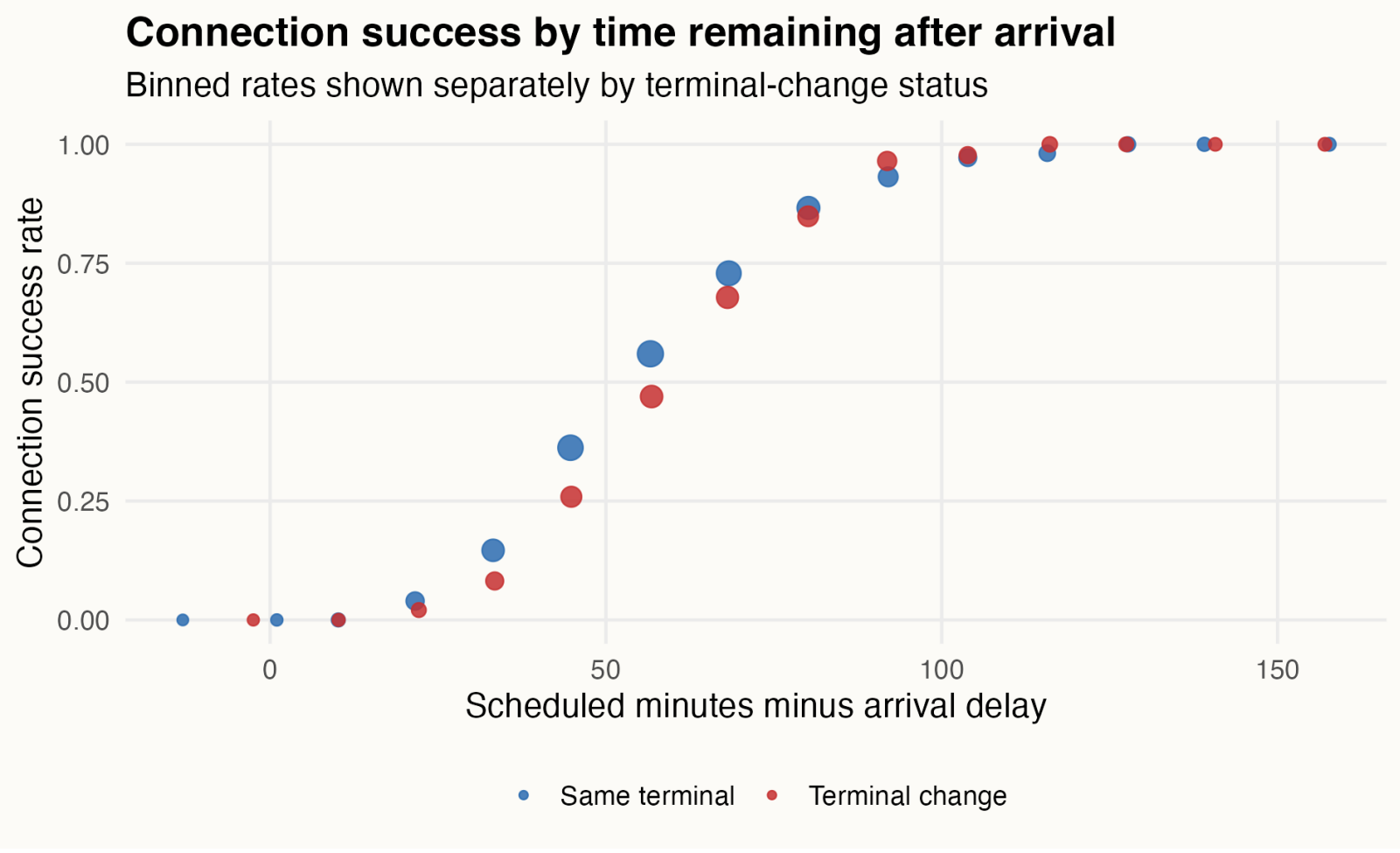
At DFW, the Skylink train makes most gate pairs reachable quickly.

At ATL, the Plane Train and parallel concourses make many transfers manageable.

At ORD or IAH, some terminal changes are more costly.

So we should not assume one universal terminal-change penalty.

# First look


Connection success by time remaining after arrival
Binned rates shown separately by terminal-change status
Connection success rate
1.00
0.75
0.50
0.25
0.00
0
50
100
150
Scheduled minutes minus arrival delay
Same terminal
Terminal change

# Main effect only

A main-effect model would say:

$$\log\left(\frac{p_i}{1-p_i}\right) = \cdots + \gamma_{\mathsf{Hub}_i} + \delta\mathsf{TerminalChange}_i.$$

This gives:

- each hub its own baseline,
- one shared terminal-change penalty.

That shared penalty is probably too blunt.

# Interaction

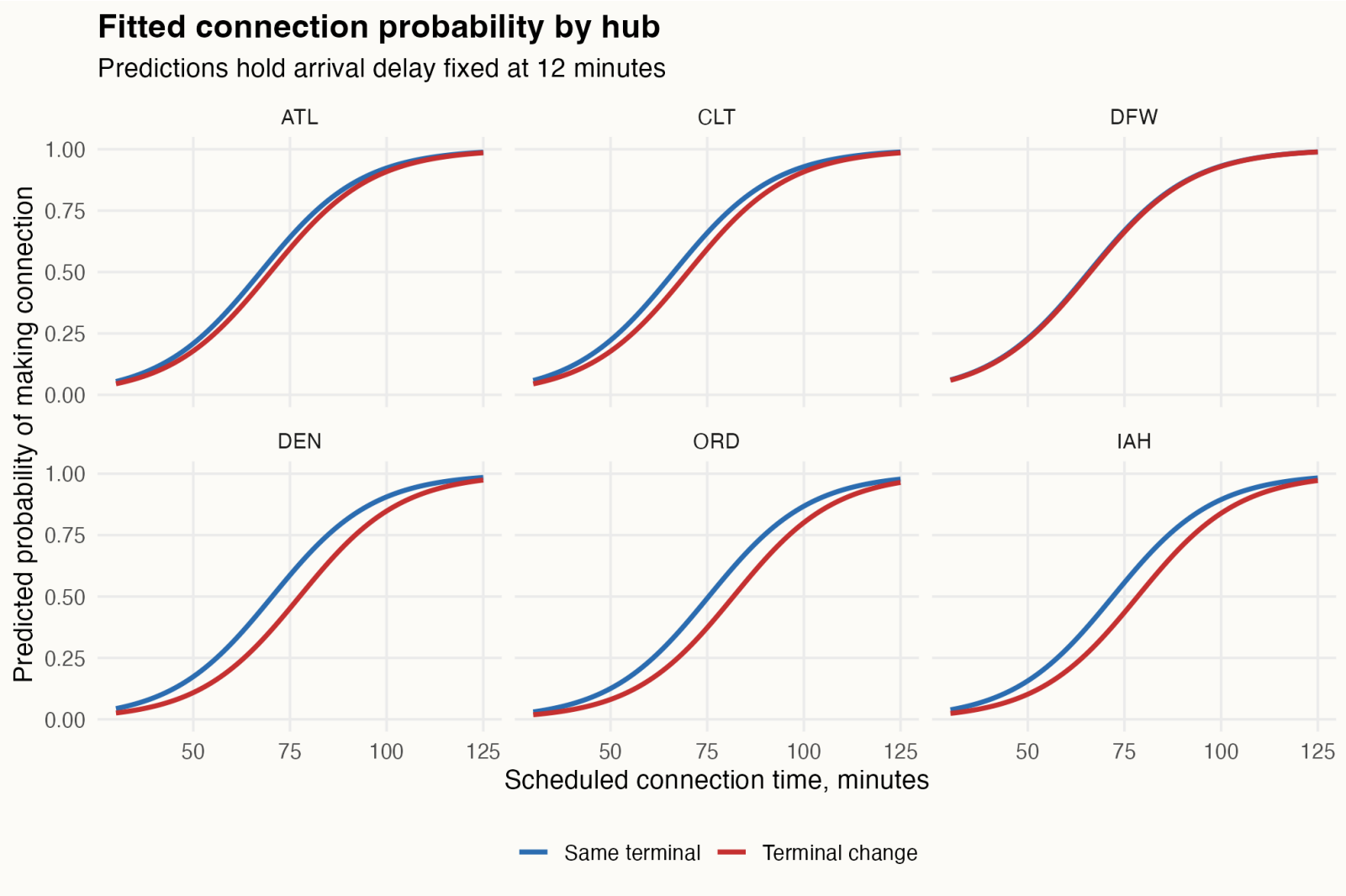
Let the terminal-change effect depend on the hub:

$$\log\left(\frac{p_i}{1-p_i}\right) = \cdots + \gamma_{\text{Hub}_i} + \delta \text{TerminalChange}_i + \theta_{\text{Hub}_i} \text{TerminalChange}_i.$$

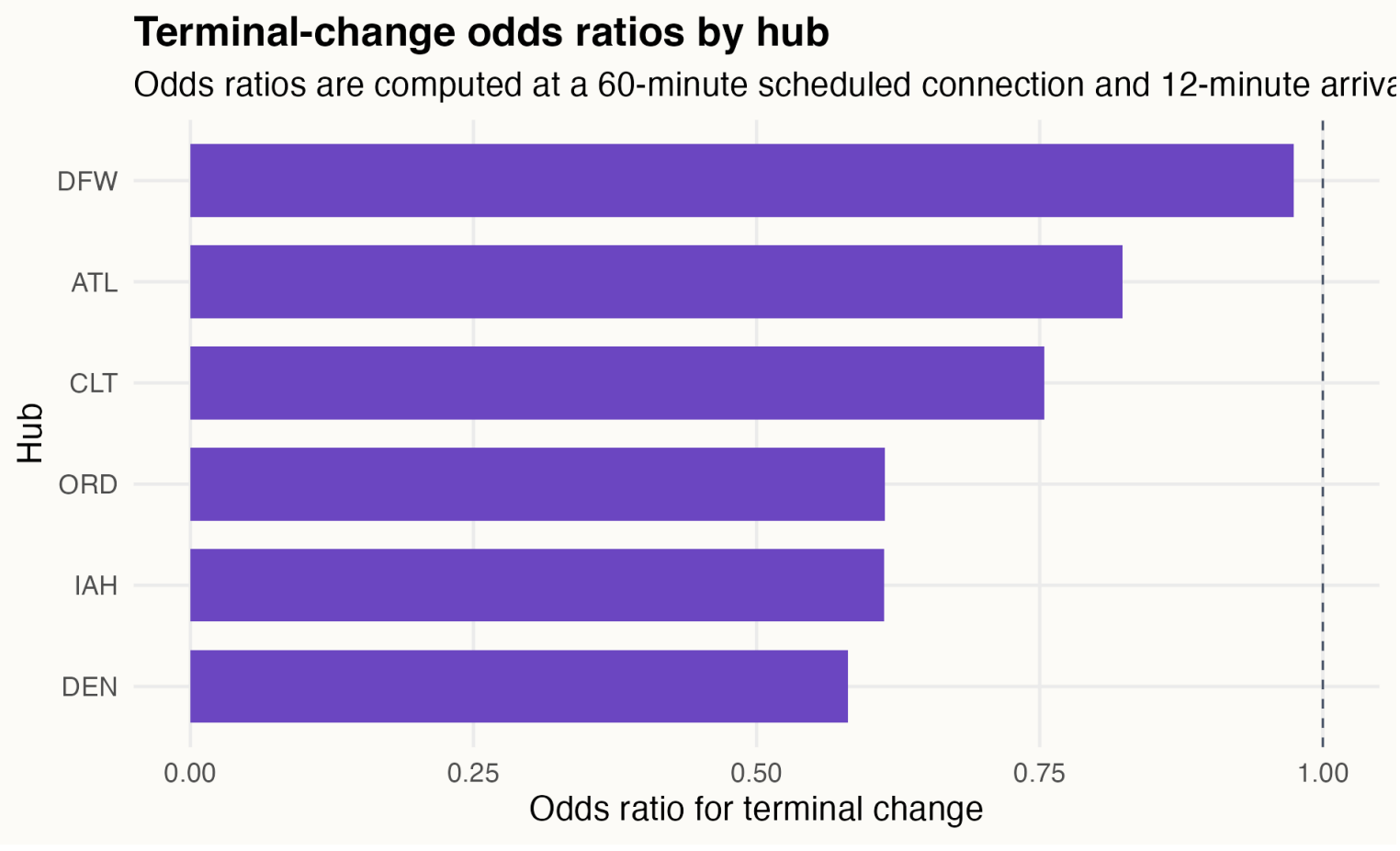
Now the terminal-change odds ratio can be different at DFW, ATL, ORD, and so on.

This is exactly the same interaction idea as before.

The response is binary, so the model is logistic.


Fitted connection probability by hub
Predictions hold arrival delay fixed at 12 minutes
ATL
CLT
DFW
DEN
ORD
IAH
Predicted probability of making connection
1.00
0.75
0.50
0.25
0.00
50
75
100
125
Scheduled connection time, minutes
Same terminal
Terminal change

# Terminal-change odds ratios



# What we learn

For a 60-minute scheduled connection with a 12-minute arrival delay:

- DFW terminal change odds ratio: about `0.97`,
- ORD terminal change odds ratio: about `0.61`,
- IAH terminal change odds ratio: about `0.61`.

So the effect is not just:

> *terminal changes are bad.*

It is:

> *terminal changes are bad in some hub layouts, and much less damaging in others.*

# The practical lesson

Logistic regression is for binary outcomes.

The model is linear on the log-odds scale:

$$\log\left(\frac{p}{1-p}\right) = \text{linear predictor.}$$

Exponentiated coefficients are odds ratios.

But the modeling grammar is familiar:

*start with the question, adjust for the right variables, and use interactions when the effect depends on context.*